

# An implementation note on HCAPO:

how a wrong  $\rho$  denominator manufactured a length runaway,  
and first evidence that the paper-correct form does not

Agentic-RL-CA project

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## Abstract

While ablating HCAPO (arXiv:2603.08754) on SearchQA (Qwen3-4B, non-thinking, 2048-token response cap), our runs reproduced a severe response-length runaway — mean length inflating  $15\times$  (89→1642 tokens/turn) with 78% of turns clipping at the cap — while accuracy stayed flat. Per-turn instrumentation of the hindsight ratio  $\rho$  traced the cause to a single line: **our port divided by the on-policy probability of the action, where the paper divides by the intra-trajectory mean of the hindsight probabilities** (Eq. 7). The wrong ratio is systematically  $< 1$  and, because only its numerator saturates with response length while the  $1/n_{tok}$  normalisation keeps growing, it *rewards verbosity*: emitting more tokens mechanically raises  $\rho$ , hence the advantage. We derive the mechanism, verify it in a controlled synthetic test (legacy  $\rho$  moves  $+0.13$  when responses lengthen  $16 \rightarrow 64$  tokens; the paper form moves  $0.0$ ), and show the corrected implementation restores every property the paper assumes:  $\rho$  centred on  $1.0$ , two-sided (the upper clip binds for the first time), with growing rather than collapsing spread. Forty-five steps into the corrected run, length and clipping are normal; the legacy runaway’s onset window (steps  $\sim 75\text{--}200$ ) has not yet been reached, so the behavioural confirmation is preliminary. All prior HCAPO numbers from this codebase (SearchQA 1.7B/4B, AlfWorld 1.7B) characterise the buggy variant, not the published method.

## 1 The two formulas

**What the paper specifies** (arXiv:2603.08754, Eq. 6–7). Per turn  $t$  of a trajectory with  $T$  turns, with  $s_{final}$  the trajectory’s last observation injected into the prompt:

$$\pi_{hind}(a_t) = \exp\left(\frac{1}{T_{temp}|a_t|} \sum_j \log \pi_\theta(y_j \mid y_{<j}, s_t, s_{final})\right) \quad (\text{Eq. 6})$$

$$\rho_t = \text{clip}\left(\pi_{hind}(a_t) / \bar{\pi}_{hind}, C_{min}, C_{max}\right), \quad \bar{\pi}_{hind} = \frac{1}{T} \sum_{k=1}^T \pi_{hind}(a_k) \quad (\text{Eq. 7})$$

The denominator is the **intra-trajectory mean over that trajectory’s turns** — the paper calls it “group-normalization across actions within the same episode”. By construction  $\rho$  is centred on  $1.0$ : pivotal turns score  $> 1$ , redundant turns  $< 1$ , and the symmetric clip  $[C_{min}, C_{max}] = [0.8, 1.2]$  is well-specified around that centre.  $\rho$  then scales a discounted return,  $Q_t^H = \rho_t \gamma^{T-1-t} R$ , which is group-normalised across the  $G$  rollouts of the prompt and added (weight  $\omega$ ) to the standard GRPO term.

**What we implemented** (gigpo/core\_hcapo.py, pre-2026-07-29):

```
mean_logratio = ((hindsight_log_probs - policy_log_probs) * mask).sum(-1) / ntok
rho = exp(mean_logratio / t_temp).clamp(0.8, 1.2)
```

```
# == pi_hind(a_t) / pi_POLICY(a_t)      <- divided by the ON-POLICY prob of the
      action
```

The numerator matches Eq. 6 exactly (per-token-normalised, temperature-sharpened hindsight log-prob). The denominator does not: instead of comparing each turn’s hindsight score against its *siblings in the same trajectory*, it compares hindsight against on-policy conditioning of the *same* tokens. That ratio measures something else entirely: *how much the injected  $s_{final}$  text perturbs the model off-distribution* — which is why, measured over 325 training steps, our  $\rho$  was **never once above 1.0** (max 0.9986; the upper clip literally never bound) while 10.5% of turns pinned at the 0.8 floor.

## 2 Why the wrong ratio rewards verbosity

Write the per-token log-ratio  $\delta_j = \log \pi(y_j | y_{<j}, s, s_{final}) - \log \pi(y_j | y_{<j}, s)$ , so `mean_logratio` =  $\frac{1}{n_{tok}} \sum_j \delta_j$ . If  $\delta_j$  had a position-independent mean,  $n_{tok}$  would cancel and length would not matter. It does not: the hindsight text perturbs the *prompt*, and both conditionings share the identical generated prefix  $y_{<j}$ , so  $\delta_j$  is large (negative) on the first tokens — where the prompt dominates the context — and decays toward 0 as the response’s own text takes over. The numerator therefore **saturates** at some constant  $C^-$  while the divisor grows:

$$\text{mean\_logratio} \approx \frac{C^-}{n_{tok}} \xrightarrow{n_{tok} \uparrow} 0^-, \quad \Rightarrow \quad \rho = \exp(\text{mean\_logratio}/T_{temp}) \rightarrow 1^-.$$

Longer responses  $\Rightarrow$  higher  $\rho \Rightarrow$  higher  $Q^H$ .

Two independent confirmations:

**(a) The training data itself.** Inverting the logged  $\rho$  to the implied *total*  $\sum_j \delta_j = T_{temp} \ln(\rho) n_{tok}$ : at step 25 ( $n_{tok}=89$ ,  $\rho=0.909$ ) the sum is  $-42.5$ ; at step 300 ( $n_{tok}=1635$ ,  $\rho=0.996$ ) it is  $-36.0$ . **Length grew 18 $\times$ ; the total hindsight effect did not grow at all** (Fig. 1, left). The entire rise of  $\rho$  is the denominator. (Order-of-magnitude arithmetic: early values are censored by the 0.8 clip floor, and inverting a batch mean through exp is Jensen-approximate.)

**(b) A controlled synthetic test.** Fix a front-loaded perturbation ( $\delta_j = -2$  on the first 8 tokens, 0 after) and vary only response length (Fig. 2): the legacy  $\rho$  climbs from 0.82 toward 1.0 as  $n_{tok}$  grows (+0.13 between 16 and 64 tokens); the paper’s  $\rho$  is **exactly length-invariant** — numerator and denominator share the same per-token-normalised functional form, so any uniform length change cancels. Note this immunity does *not* depend on the front-loading diagnosis being right, which is a good property.

**How a  $\rho$  bias becomes a length ratchet.**  $Q^H$  feeds a group normalisation across the  $G=5$  rollouts of each prompt, which is invariant to uniform scaling — so “everyone gets longer, everyone gains” cancels. The gradient pressure is *relative*: among sibling rollouts of the same prompt, the **longer rollout has higher  $\rho$  than its siblings**, out-scores them, and is reinforced. Each iteration the whole group lengthens, but the relative pressure persists — a ratchet. This predicts both observed dynamics (Fig. 3): a *phase transition* rather than smooth drift (legacy A: 551  $\rightarrow$  1371 tokens in 25 steps once the loop won), and *self-termination* as  $\rho$  approaches its ceiling and the across-sibling spread compresses (legacy A:  $\rho$  std 0.055  $\rightarrow$  0.038; length plateaus at  $\sim 1635$  against the 2048 cap, clip 0.77–0.78, *above* the 0.754 that accompanied the 1.7B thinking-mode collapse). Halving the hindsight weight ( $\omega=0.5$ , legacy B) shrinks the step size along the same gradient — it **delays the runaway by  $\sim 125$  steps but does not prevent it**. Throughout, accuracy stayed flat ( $\approx 0.39$  macro-EM): the verbosity was *waste*, not fuel.

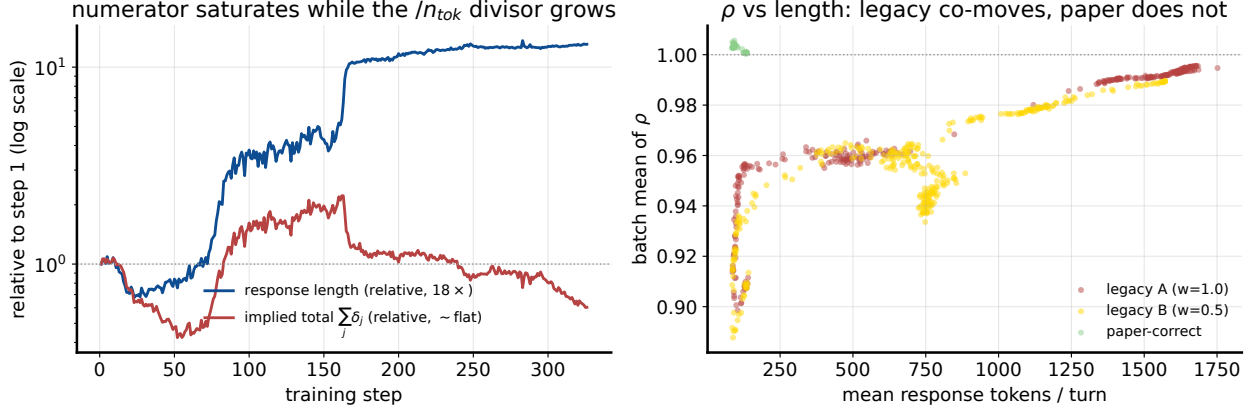


Figure 1: **The dilution mechanism, measured.** *Left:* in the legacy  $\omega=1$  run, the implied total hindsight log-ratio  $\sum_j \delta_j$  (red) is  $\sim$ flat while response length (blue) grows  $18\times$  —  $\rho$  rises purely because the  $1/n_{tok}$  divisor grows. *Right:* batch-mean  $\rho$  against response length: the two legacy runs trace a tight monotone curve toward 1.0; the paper-correct run sits at  $\rho \approx 1$  independent of length.

### 3 The corrected implementation

`gigpo/core_hcapo.py` now implements Eq. 6–7 as the default (`rho_denominator="intra_traj_mean"`; the buggy path is retained under `"on_policy"` solely to reproduce the withdrawn runs):

```
# Eq. 6: per-token-normalised, temperature-sharpened hindsight log-prob
log_pi_hind = (hindsight_log_probs * mask).sum(-1) / ntok / t_temp
# Eq. 7: self-normalise by the mean over the TURNS of the same trajectory
for tu in unique(traj_uid):
    idx = rows_of(tu); rep = one_row_per_distinct_turn(idx)
    xm = log_pi_hind[rep].max() # numerical stability shift
    rho[idx] = exp(log_pi_hind[idx] - xm) / mean(exp(log_pi_hind[rep] - xm))
rho = clip(rho, 0.8, 1.2)
```

The policy log-probs no longer enter  $\rho$  at all, matching the paper. Unit tests: (i) the synthetic length test above (legacy  $+0.1325$ , paper  $0.0000$ ); (ii)  $\rho$  centred on 1.0 with two-sided spread on toy trajectories.

### 4 Initial observations: why the corrected form does not explode

**Structural reason.** The ratchet needs a way for a rollout to raise its  $\rho$  by being globally more verbose than its siblings. Under Eq. 7 it has none:  $\rho_t$  compares turn  $t$  only against *the same trajectory's other turns*, and numerator and denominator carry the same  $1/|a|$  normalisation, so uniform verbosity cancels identically. What remains is a bounded, *relative* preference between turns of one trajectory (a longer turn can score somewhat above its shorter siblings), clipped to  $[0.8, 1.2]$  and non-compounding — there is no cross-rollout length gradient for group normalisation to amplify.

**Empirical, first 45 steps** (single seed; all values from the run's per-step instrumentation):

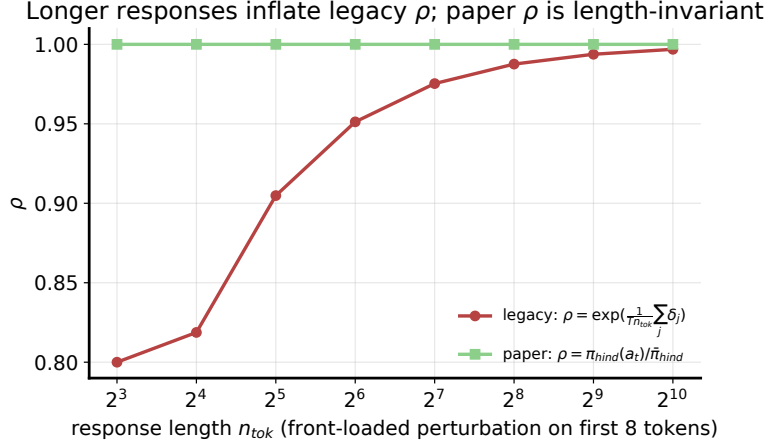


Figure 2: **Controlled test of the mechanism.** Front-loaded perturbation ( $\delta = -2$  on the first 8 tokens), response length varied, nothing else. Legacy  $\rho$  (red) inflates with length; the paper’s self-normalised  $\rho$  (green) is exactly 1 at every length.

|                                                | legacy (withdrawn), step 1 | paper-correct, step 1 |
|------------------------------------------------|----------------------------|-----------------------|
| $\rho$ mean                                    | 0.9078                     | 1.0006                |
| $\rho$ min / max                               | 0.800 / 0.9986             | 0.800 / 1.2000        |
| frac at lower clip                             | 0.105                      | 0.023                 |
| frac at upper clip                             | 0.000                      | 0.014                 |
| $ \text{macro}  :  \text{micro} $ (adv. terms) | 1 : 1.40                   | 1 : 1.14              |

Every property the paper assumes is restored:  $\rho$  is centred on 1.0 and **two-sided** — **the upper clip binds for the first time in this codebase** (1.4% of turns), i.e. some turns are now scored as *pivotal*, which is the entire point of hindsight credit and was structurally impossible under the legacy ratio. The spread is *growing* (std 0.073  $\rightarrow$  0.143 over 45 steps) where the legacy spread collapsed (0.055  $\rightarrow$  0.038): the signal differentiates turns instead of saturating (Fig. 4). Behaviourally, at step 25: macro-EM 0.352, clip 0.001, response length 91 tokens — indistinguishable from GRPO at the same point, as expected.

**What this does not yet show.** The legacy runaway ignited in the window steps  $\sim 75$ –200 (171  $\rightarrow$  487  $\rightarrow$  1371 tokens). The corrected run has not reached it; “no explosion” is therefore a structural argument plus 45 clean steps, not yet a completed demonstration. The decisive within-batch test — per-sample  $\text{corr}(\rho, n_{tok})$ , predicted  $\gg 0$  for the legacy form and  $\approx 0$  for Eq. 7 — is queued for the next instrumentation pass. This note will be updated at steps 100/200/500.

## 5 Record

Withdrawn: both legacy 4B runs (killed at steps  $\sim 324/330$ ), and all earlier HCAPO numbers from this codebase — SearchQA 1.7B (0.357 @ clip 0.754), 4B legacy arms ( $\approx 0.39$ ), AlfWorld 1.7B (0.791) — as characterisations of the *published* method (they remain valid measurements of the buggy variant). The full timestamped trail (pre-registration, amendments, retraction of two premature trend calls, and this correction) is in docs/reports/2026-07-21\_4b\_horizon\_eval/results/hcapo\_wave3\_preregistration.md. Provenance caveat: Eq. 6–7 were transcribed from the arXiv HTML via automated fetch (re-queried twice, quoted verbatim); confirm against the PDF before external use. Rebuild: assets/build\_figures.py (parses the three training logs directly), then tectonic report.tex.

Legacy HCAPO runs away in length; the paper-correct run (so far) does not

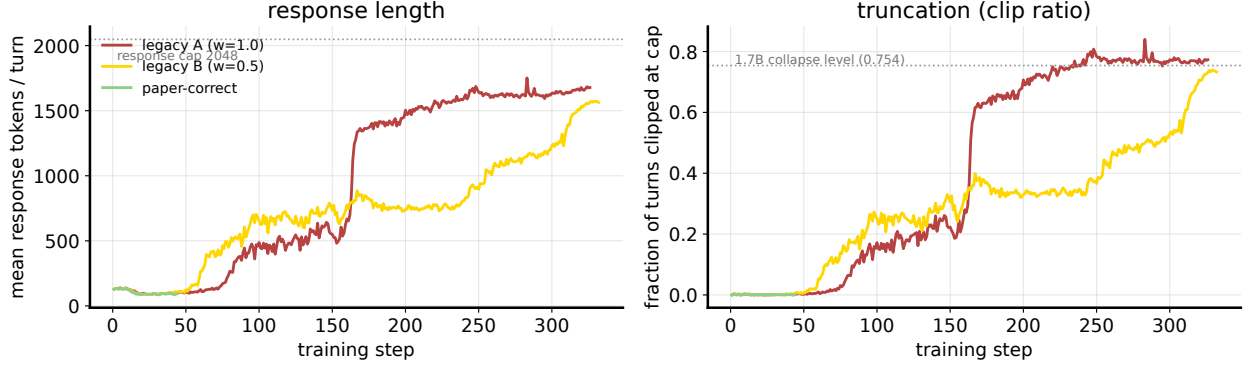


Figure 3: **The runaway, and its absence so far under the fix.** Mean response length (left) and cap-clip fraction (right) per training step. Legacy  $\omega=1$  (red) undergoes a phase transition at  $\sim$ step 160 and saturates against the 2048 cap;  $\omega=0.5$  (orange) follows the same path  $\sim$ 125 steps later; the paper-correct run (green, 45 steps so far) is flat — though it has not yet reached the window where the legacy runs took off.

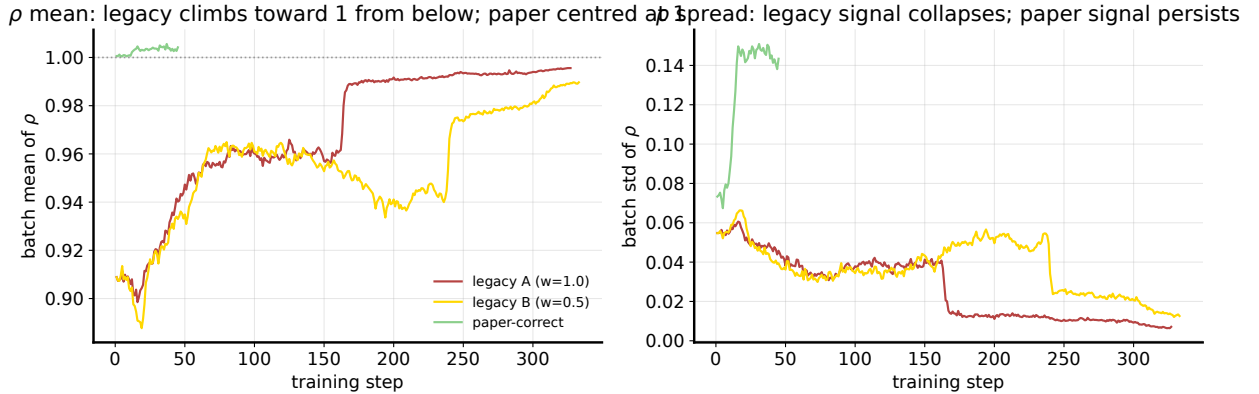


Figure 4:  **$\rho$  statistics per training step.** *Left:* legacy  $\rho$  (red/orange) starts at 0.91 and is crushed toward 1.0 from below as length inflates; the paper-correct run (green) sits at 1.0 from step one. *Right:* legacy spread collapses as  $\rho$  saturates; the corrected spread grows — the hindsight signal discriminates between turns rather than encoding response length.